

Standard

Performance-Based Product Failure Mode, Effects and Criticality Analysis (FMECA) Requirements

AIAA standards are copyrighted by the American Institute of Aeronautics and Astronautics (AIAA), 1801 Alexander Bell Drive, Reston, VA 20191-4344 USA. All rights reserved.

AIAA grants you a license as follows: The right to download an electronic file of this AIAA standard for storage on one computer for purposes of viewing, and/or printing one copy of the AIAA standard for individual use. Neither the electronic file nor the hard copy print may be reproduced in any way. In addition, the electronic file may not be distributed elsewhere over computer networks or otherwise. The hard copy print may only be distributed to other employees for their internal use within your organization.



American National Standard

Performance-Based Product Failure Mode, Effects and Criticality Analysis (FMECA) Requirements

Sponsored by

American Institute of Aeronautics and Astronautics

Approved 17 November 2008

American National Standards Institute

Abstract

This standard provides the basis for developing the analysis of failure modes, their effects, and criticality in the context of individual products along with the known performance of their elements. The requirements for contractors, the planning and reporting needs, along with the analytical tools are established. The linkage of this standard to the other standards in the new family of performance-based reliability and maintainability standards is described, and all of the keywords for use in automating the Product FMECA process are provided.

Library of Congress cataloging-in-publication data on file

Published by
American Institute of Aeronautics and Astronautics
1801 Alexander Bell Drive, Reston, VA 20191

Copyright © 2009 American Institute of Aeronautics and
Astronautics
All rights reserved

No part of this publication may be reproduced in any form, in an electronic retrieval system
or otherwise, without prior written permission of the publisher.

Printed in the United States of America

Contents

Foreword.....	v
1 Scope.....	1
1.1 Purpose.....	1
1.2 Application	2
2 Applicable Documents	2
2.1 Normative References	2
2.2 Relationship To Other S-102 Standards	3
3 Vocabulary.....	4
3.1 Acronyms and Abbreviated Terms.....	4
3.2 Terms and Definitions	4
4 General Requirements.....	6
4.1 Contractor Responsibility	6
4.2 Planning.....	7
4.3 Product FMECA Report	7
5 Detailed Requirements.....	7
5.1 Define Product FMECA Ground Rules.....	7
5.2 System Design Data Collection.....	8
5.3 Failure Mode and Effects Analysis.....	8
5.4 Criticality Analysis	10
5.5 Failure Detection Analysis	10
5.6 Failure Isolation Analysis	10
5.7 Detection Risk Priority Analyses	10
5.8 Reliability, System Safety, and Maintainability Critical Item Analyses	10
5.9 Failure Compensation Method Analysis	10
5.10 Integrated Product FMECA/Hazards Analysis Database.....	11
5.11 Data Exchange Between Product FMECA Process And Other Project Functions.....	12
5.12 Product FMECA Process Performance Evaluation.....	12
5.13 Lessons Learned.....	14
5.14 Structured Review	15
Annex A AIAA S-102 Document Tree (Normative).....	18
Annex B AIAA S-102 Product FMECA Capability Level Requirements (Normative).....	19
Annex C AIAA S-102 Product FMECA Keyword Data Element Descriptions (Normative)	23
Figures	
Figure 1 — Failure Process Flow	8
Figure 2 — Product FMECA Dataset Evaluation.....	16

Figure 3 — Failure Compensation Method Evaluation	17
---	----

Tables

Table 1 — AIAA S-102 Failure Severity Classification.....	9
---	---

Table 2 — Product FMECA Dataset Maturity Rating Criteria.....	13
---	----

Table 3 — Failure Compensation Method Maturity Rating Criteria.....	14
---	----

Foreword

A performance-based Product Failure Mode, Effects, and Criticality Analysis (FMECA) standard has been developed to aid organizations in assuring that their product FMECA tasking presents a “value-added” contribution to the product development effort. The need for such a standard arises from the absence of an accepted methodology for assessing the capability of reliability and maintainability (R&M) programs. The ISO 9000 series standards presently guide organizations in establishing practices for consistently producing superior products, but only within the discipline of quality assurance. The ISO 9000 series of standards do not define processes and practices for achieving high-reliability products. Because the criteria for quality and reliability differ, an organization that conforms to stringent workmanship specifications and produces high-quality products through participation in the ISO 9000 certification process may still lack the capability to consistently deliver high-reliability products. Hence, this product FMECA standard was developed as part of a suite of AIAA S-102 performance-based R&M standards for quantifying the results of, and improving the performance of, R&M programs.

The AIAA S-102 performance-based R&M standards document tree (Annex A) includes 35 standards that provide criteria for rating the capability of R&M practices. They represent proven approaches for planning and implementing the product life cycle R&M program. The S-102 R&M capability-rating criteria allow organizations to:

- specify a level of R&M program performance,
- plan the activities necessary to achieve a level of R&M program performance,
- appraise the performance an R&M program or individual practice, and
- identify the activities necessary to improve the performance of an R&M program or individual practice.

The S-102 R&M capability-rating criteria (Annex B in all S-102 standards) are intended to aid organizations in assuring their R&M programs are a “value-added” contribution to the product development effort. *There is no intent to prescribe a universal methodology for quantifying the evaluation or improvement of R&M programs or individual practices.* The S-102 R&M capability-rating criteria reflect the collective body of knowledge of the S-102 Working Group, which was chartered by the AIAA Standards Executive Council to develop and approve the S-102 standards. The S-102 Working Group is comprised of R&M experts that represent the government and industry sectors affected by the S-102 standards.

This standard establishes uniform requirements for a performance-based product FMECA. The principles of the product FMECA as promoted in a performance-based approach can be learned from this standard alone, but its proper use requires careful planning, for which the prerequisite is, understanding associated S-102 documents and identifying the desired R&M data products in the systems engineering process. What distinguishes this standard from all past and present product FMECA standards is the following:

- It provides consistent criteria for rating the “capability” of the product FMECA process.
- It provides consistent criteria for rating the “maturity” of the product FMECA data.
- It calls for the use of knowledge-based approaches to identify, analyze, and manage failure modes that pose unacceptable risk.
- For a Capability Level 3 or above FMECA process, it calls for the collection and review of existing lessons learned, and the generation and formal approval of new lessons learned.
- For a Capability Level 4 or above product FMECA process, it calls for the use of predefined R&M data parameters to facilitate the exchange of product FMECA data among computer-aided analysis tools and other project databases.

At the time of approval the members of the AIAA Performance-Based Reliability and Maintainability Standards Working Group were:

Tyrone Jackson (Chair)	SRS Technologies
Lily Lau	The Aerospace Corporation
David Oberhettinger	NASA Jet Propulsion Laboratory
Walt Willing	Northrop Grumman Electronic Systems
Steve Harbater	Northrop Grumman Integrated Systems
Alazel Jackson	Raytheon Space and Airborne Systems
Jan Swider	Pratt & Whitney Rocketdyne, Inc.
Dev Raheja	Design for Competitiveness
Jeff Merrick	Merrick Consulting
Ken Gibson	Boeing Space and Intelligence Systems
James French	RMS Partnership
Ari Jain	Reliability Consultant
Dawson Coblin	Lockheed Martin Space Systems
Terry Hardy	Federal Aviation Administration

The above consensus body approved this document in June 2006.

The AIAA Standards Executive Council (Mr. Amr ElSawy, Chairman) accepted the document for publication in July 2008.

The AIAA Standards Procedures dictates that all approved Standards, Recommended Practices, and Guides are advisory only. Their use by anyone engaged in industry or trade is entirely voluntary. There is no agreement to adhere to any AIAA standards publication and no commitment to conform to or be guided by standards reports. In formulating, revising, and approving standards publications, the committees on standards will not consider patents that may apply to the subject matter. Prospective users of the publications are responsible for protecting themselves against liability for infringement of patents or copyright or both.

1 Scope

This standard establishes uniform requirements and criteria for a performance-based, product Failure Mode, Effects and Criticality Analysis (FMECA). The performance-based aspect of this standard requires that the organization's FMECA capability be rated according to predetermined criteria for process capability and data maturity.

1.1 Purpose

The primary purpose of the product FMECA is to collect and evaluate the necessary system design information to identify and eliminate or control all failure modes that pose unacceptable risk. Depending on how it is performed, the FMECA can be used for many purposes, but its most important use in systems engineering is to aid the improvement of design reliability before the product is manufactured. For this standard, a failure mode is defined as the consequence of the mechanism through which the failure occurs, i.e., the manner by which the failure is observed. Accordingly, a failure mode that poses unacceptable risk is a failure mode whose effect, either singularly or in combination with other failure mode effects, violates a system design requirement or goal. The FMECA is a systematic methodology that is widely used to evaluate the effects on systems and interfaces caused by the failure modes of functional or physical components, to estimate the criticality of each failure mode in terms of its end-effects, and to evaluate the appropriate failure compensation methods for mission critical failure modes. The S-102 Product FMECA is comprised of three components:

- failure mode and effects analysis (FMEA),
- criticality analysis (CA) and risk priority analysis (RPA), and
- critical item analysis (CIA) and failure compensation analysis (FCA).

The FMEA answers system failure questions regarding the *what's*, *how's*, *where's*, *when's*, and *why's*. The CA and the RPA answer system failure questions regarding *relative significance*. The CIA and the FCA answer system failure questions regarding *failure mitigation*. The minimum activities that constitute the baseline practice for the product FMECA are the following:

- timely establishment of the requirements and analytical ground-rules for the product FMECA;
- timely establishment of product FMECA technical performance metrics (TPM);
- timely collection and evaluation of the necessary system design information to identify and eliminate or control all inherent, singular hardware (H/W) failure modes that pose unacceptable risk across the system life cycle;
- timely creation of a product FMECA dataset for each postulated system failure mode;
- timely application of reliability critical item (RCI) selection criteria and the evaluation of RCI failure modes;
- timely calculation of the criticality number (CN) and detection risk priority number (RPN);
- timely identification of scheduled project activities that are intended to help reduce the failure likelihood or manage the consequences of the failure;
- timely identification and documentation of all non-credible Severity Classification 3 (i.e., Marginal) or higher inherent, singular H/W failure modes;
- timely validation of each product FMECA dataset; and
- timely development and documentation of the product FMECA report.

1.2 Application

This standard applies to acquisitions for the design, development, fabrication, test, and operation of commercial, civil, and military systems, equipment, and associated computer programs. This standard provides criteria for rating the capability of an acquisition project's product FMECA practices, and assigning the practice to one of five predetermined levels of performance capability. The capability criteria provide the logical sequence of activities for improving the effectiveness of an existing product FMECA practice by advancing in stages. Organizations may evaluate their existing product FMECA practices against these minimum acceptable criteria to identify the features that need to be added to improve their existing product FMECA practices.

This standard also applies to the integration of the product FMECA/hazards analysis database with a project R&M database system. However, specification of this standard shall not be construed as a requirement for the contractor to use a particular computer-aided design (CAD) system. Rather, contractors may use the computerized tools of their choosing, given that those tools are validated to process input data and generate output data in formats that are compatible with the data definitions in this standard.

2 Applicable Documents

2.1 Normative References

The following reference documents of the issue in effect on the date of invitation for bid or request for proposal form a part of this standard to the extent specified:

- 1) AIAA S-102.0.1 (Draft) Performance-Based R&M Program General Requirements

AIAA S-102.1 Performance-Based R&M Management

- 2) AIAA S-102.1.1 (Draft) R&M Program Planning Requirements
- 3) AIAA S-102.1.6 (Draft) R&M Risk Management Requirements
- 4) AIAA S-102.1.7 (Draft) Project R&M Database System Requirements

AIAA S-102.2 Performance-Based R&M Engineering and Analysis

- 5) AIAA S-102.2.1 (Draft) Functional Dependency and Architectural Modeling Requirements
- 6) AIAA S-102.2.2 System Reliability Modeling Requirements
- 7) AIAA S-102.2.3 (Draft) Component Reliability Predictions Requirements
- 8) AIAA S-102.2.5 Sneak Circuit Analysis Requirements
- 9) AIAA S-102.2.6 (Draft) Design Concern and Drawing Error Analysis Requirements
- 10) AIAA S-102.2.7 (Draft) Parts Stress Derating Analysis Requirements
- 11) AIAA S-102.2.8 (Draft) Worst Case Analysis Requirements
- 12) AIAA S-102.2.9 (Draft) Human Error Predictions Requirements
- 13) AIAA S-102.2.11 Anomaly Detection and Response Analysis Requirements
- 14) AIAA S-102.2.12 (Draft) Maintainability Predictions Requirements
- 15) AIAA S-102.2.13 (Draft) Operational Dependability and Availability Modeling Requirements
- 16) AIAA S-102.2.14 (Draft) Hazard Analysis Requirements

- 17) AIAA S-102.2.15 (Draft) Software Reliability Predictions Requirements
- 18) AIAA S-102.2.16 (Draft) Process Failure Mode, Effects and Criticality Analysis Requirements
- 19) AIAA S-102.2.20 (Draft) R&M Similarity and Allocations Analysis Requirements
- 20) AIAA S-102.2.21 (Draft) Parts Engineering and Analysis Requirements
- 21) AIAA S-102.2.22 (Draft) Stress and Damage Simulation Analysis Requirements

AIAA S-102.3 Performance-Based R&M Testing

- 22) AIAA S-102.3.2 (Draft) Reliability Development Growth Testing Requirements

AIAA S-102 Performance-Based R&M Guidance

- 23) AIAA S-102-Annex B.4.2 (Draft) Criticality Analysis Guide
- 24) AIAA S-102-Annex B.5 (Draft) Sneak Circuit Analysis Guide
- 25) AIAA S-102-Annex B.6 (Draft) Design Concern and Drawing Error Analysis Guide

2.2 Relationship To Other S-102 Standards

This standard is one of 35 standards in the AIAA S-102 standards document tree. The S-102 standards define rated sets of activities that aid the identification, assessment¹, and management of R&M risks for high unit-value systems. For S-102 standards, the performance-based R&M program is defined as the set of activities that address product R&M risk factors at one or more predetermined capability levels.

In the schema for the S-102 performance-based R&M program (see AIAA S-102.1.1 (Draft)), the product FMECA supports several R&M practices. Accordingly, this standard identifies the product FMECA data products that are inputs for performance-based R&M practices defined by the following S-102 standards:

AIAA S-102.1.7 (Draft)	AIAA S-102.2.7 (Draft)	AIAA S-102.2.14 (Draft)
AIAA S-102.2.1 (Draft)	AIAA S-102.2.8 (Draft)	AIAA S-102.2.15 (Draft)
AIAA S-102.2.3 (Draft)	AIAA S-102.2.9 (Draft)	AIAA S-102.2.16 (Draft)
AIAA S-102.2.5	AIAA S-102.2.11	AIAA S-102.2.20 (Draft)
AIAA S-102.2.6 (Draft)	AIAA S-102.2.12 (Draft)	AIAA S-102.2.21 (Draft)

Also, this standard identifies the Product FMECA inputs that are data products of performance-based R&M practices defined by the following S-102 standards:

AIAA S-102.1.6 (Draft)	AIAA S-102.2.22 (Draft)
AIAA S-102.2.2	AIAA S-102.3.2 (Draft)
AIAA S-102.2.13 (Draft)	

¹ Assessment as used here includes qualification and quantification, and numerical evaluation and ranking of R&M risks.

3 Vocabulary

3.1 Acronyms and Abbreviated Terms

CA	Criticality Analysis
CN	Criticality Number
FMEA	Failure Mode and Effects Analysis
FMECA	Failure Mode, Effects, and Criticality Analysis
H/W	Hardware
ILS	Integrated Logistics Support
LLAA	Lessons Learned Approval Authority
PMP	Parts, Materials and Processes
QA	Quality Assurance
R&M	Reliability and Maintainability
RPA	Risk Priority Analysis
RPN	Risk Priority Number
SPFM	Single Point Failure Mode
S/W	Software

3.2 Terms and Definitions

acquisition activity

activity (government, contractor, or subcontractor) that levies product FMECA requirements on another activity through a contract or other document

ambiguity group

set of components that, if considered potentially faulty, does not give a unique solution in fault location

baseline practice

minimum set of activities that constitute an AIAA S-102 Capability Level 1 R&M practice

capability level

measure of the ability of an R&M practice, as specified by a set of activities, to address the pertinent R&M needs of a high unit-value systems engineering process

capability level growth

improvement in the ability of a task to support the systems engineering objectives due to an increase in resources, engineering data processed, or number of other tasks supported

credible failure mode

failure mode with a probability of occurrence greater than $1.0E-6$, 0.000001, or one in a million

criticality analysis

grouping of probability of occurrences for failure modes according the Severity Classification

NOTE Each failure mode probability of occurrence represents a criticality number (CN).

dataset

analysis output that consists of a group of related data elements

detection RPA

grouping of the products of probability of occurrences and detection success rate for failure modes according to Severity Classification

NOTE In this case, the RPN is the product of the CN and detection success rate.

failure

termination of the ability of a unit to perform its required function

NOTE A fault may cause a failure.

failure mode

consequence of the mechanism through which a failure occurs, or the manner by which a failure is observed

fault²

(1) [Software Reliability] a manifestation of an error in software; (2) [Hardware Reliability] any undesired state of a component or system; (3) [Components] a defect or flaw in a hardware or software component

NOTE (1) A fault may cause a failure.

NOTE (2) A fault does not necessarily require failure.

FMECA dataset

set of data that describes a specific failure mode and its effects, root cause, detection method, isolation method, severity classification, criticality number, detection RPN, isolation RPN if applicable, and compensation method

independent verification

provision of sufficient information for an organization or individual to obtain the same results as the analysts when redoing the analysis

inherent failure mode

result of characteristic weaknesses related to specified design and materials of an item

induced failure mode

result of uncharacteristic weaknesses related to avoidable influences on one or more specified design or materials of an item

maturity level

measure of the degree of accuracy of a data product, as developed using a specified set of input data, in relation to what is considered the best achievable results

mission assurance

a top-down, comprehensive, risk management process that is performed over the life cycle of a high unit-value system to identify, evaluate, and mitigate or control all potential hazards and failures that pose an unacceptable risk to mission success

NOTE Potential, damaging-threatening hazards and mission-impacting failures may be caused by requirements, developmental activities, handling methods, environmental conditions, physical interactions, functional characteristics, or operator actions.

² Definition source: IEEE 100, *The Authoritative Dictionary of IEEE Standards Terms*

non-credible failure mode

failure mode with a probability of occurrence less than $1.0E-6$, 0.000001, or one in a million

performance-based R&M program

a set of activities that address product R&M risk factors at one or more predetermined capability levels

process-based lesson learned

important information created, documented, and retrieved according to a process or procedure descriptor

product-based lesson learned

important information created, documented, and retrieved according to a system or device life cycle specific functional or physical descriptor

prognostics

identifying conditions or clues that indicate imminent failure prior to its occurrence

tailoring

process by which the individual requirements (tasks, sections, paragraphs, words, phrases, or sentences) of a standard are evaluated to determine the extent to which each requirement is most suited for a specific system acquisition and the modification of these requirements, where necessary, to ensure that each tailored document invokes only the minimum needs of the customer

timely

performance of a task, subtask, or effort when planning and execution results in the output being provided with sufficient time for management, if need be, to identify and implement cost-effective action

EXAMPLE Action that avoids or minimizes schedule delays and cost increases) for achieving, preserving, or verifying system R&M requirements.

validation

the act of determining that a product or process, as constituted, will fulfill its desired purpose

verification

the process of assuring that a product or process, as constituted, complies with the requirements specified for it

4 General Requirements

4.1 Contractor Responsibility

The contractor shall provide the necessary training and tools to allow the product FMECA process to be cost-effectively applied as needed in a project. The contractor and his subcontractors shall plan and implement a product FMECA process. The FMECA shall evaluate all possible failure modes, their effects on mission success, and the method of failure detection. Verifiable failure compensation methods³ shall be identified for each failure mode that has unacceptable risk or rationale provided for taking no action. The timing of the analysis shall allow needed reliability improvements to be identified and implemented in a timely manner. If a Capability Level 3 System FMECA process is required, the FMECA process shall support the efforts of the Lessons Learned Approval Authority (LLAA) (e.g., Lessons Learned Committee) to identify lessons learned associated with design deficiency trends, and shall assist in infusing the lessons learned recommendations throughout the project. If a Capability Level 4 or above FMECA process is required, the data products shall be prefaced with one or more of the keyword data element descriptions (DED) listed in Annex C.

³ Failure compensation methods may include special design features, analyses, inspections, tests, controls, instructions, or operational corrective actions, such as, replacement, reconfiguration, repair, or preventive maintenance.

4.2 Planning

The contractor's product FMECA process plan shall be developed, documented, and flowed down, as appropriate, as an integral part of the R&M program plan. The contractor shall document the approved methods for implementing the product FMECA process and reference the applicable document number(s) in the plan. The product FMECA process shall be planned to support timely and cost-effective implementation of the R&M program⁴, which includes timely support of other project activities while precluding duplication of effort with these activities. The R&M program activities that the FMECA processes support shall include, but not limited to, subcontractor and supplier monitoring and control, the failure reporting, analysis & corrective action system (FRACAS), R&M risk management, system reliability modeling, and reliability development growth testing (RDGT). The non-R&M program activities that the FMECA processes support shall include, but not be limited to, test planning, quality assurance (QA), human engineering, system safety, configuration management, integrated logistics support (ILS), and parts, materials, and processes (PMP) control.

The minimum tasks that shall be prescribed in the product FMECA plan shall be based on the required capability level, as defined in Annex B. The FMEA process shall be described in terms of a specified capability that includes the Annex B activities peculiar to the specified capability level plus the activities peculiar to each lower capability level. The contractor shall consider the applicability of capability-level growth over the life cycle of a project when planning the product FMECA process.

4.3 Product FMECA Report

The contractor shall document the product FMECA as it is being performed and provide the most current product FMECA report to the acquisition activity upon request. At all times, the product FMECA report shall reflect the current state of the system design to the greatest extent practical. If a Capability Level 3 or higher product FMECA is required, the product FMECA report shall be generated from the Product FMECA/hazards analysis database.

5 Detailed Requirements

The following detailed requirements pertain to the performance-based, product FMECA tasks defined in Annex B.

5.1 Define Product FMECA Ground Rules

The contractor shall define the product FMECA ground rules that include:

- a) selecting the FMECA data sources and data processing methods⁵;
- b) describing how consistency will be achieved among independently performed FMECA;
- c) defining the procedures and nomenclature by which failure modes and effects will be documented; and
- d) describing the extent that effects due to external system interfaces and influences, e.g., human errors, operating environments, etc. will be identified.

⁴ AIAA S-102.1.1, "Performance-Based R&M Program Planning Requirements," calls for the R&M program plan to support timely and cost-effective implementation of the integrated master plan (IMP).

⁵ This is a process validation activity when it includes evaluation of the appropriateness of the product FMECA process prior to its use.

5.2 System Design Data Collection

Prior to beginning the evaluation of system failure modes, the contractor shall collect system design information for all possible inherent and induced, functional and physical, and singular, sequential, and combinational failure modes that fall within analytical ground rules to be specified by the contractor (see Reference 5). The system design information shall include all system levels, mission phases, and environments, and all normal, degraded, and contingency system modes that are applicable to each mission phase. If a Capability Level 3 or higher product FMECA is required, this information shall be entered in the product FMECA/hazards analysis database to allow cross-referencing failure modes against official design drawings. The contractor's analytical ground rules shall include the identification of all functional and physical component failure modes (including software module and physical input/output anomalies) that can lead to system failure or degraded operation, their effects on mission success (i.e., Severity Classification), and their root cause(s) when corrective action is needed to meet requirements.

5.3 Failure Mode and Effects Analysis

The contractor shall collect sufficient system design information to perform the failure mode and effects analysis (FMEA), i.e., identify and evaluate (within ground rules to be specified by the contractor) all credible and non-credible failure modes in the system design. Figure 1 illustrates the generalized failure flow process from the lowest to the highest system indenture level. Depending on the product FMECA capability level that is required, the contractor shall use one or more of the following approaches to evaluate the system's failure flow processes:

- Perform bottom-up analysis to identify failure effects. Start with failure modes at selected components indenture levels and proceed through increasing indenture levels until the end-effects are identified.
- Perform top-down analysis to identify failure causes as necessary to evaluate compensating methods for critical failure modes. Start with anomalous effects at selected components indenture levels and proceed through decreasing indenture levels until the root causes are identified.
- Perform event sequence analysis to identify failure mechanisms as necessary to evaluate compensating methods for critical failure modes. Start with causal factors at selected components indenture levels and proceed forward until the failure mechanisms are identified.

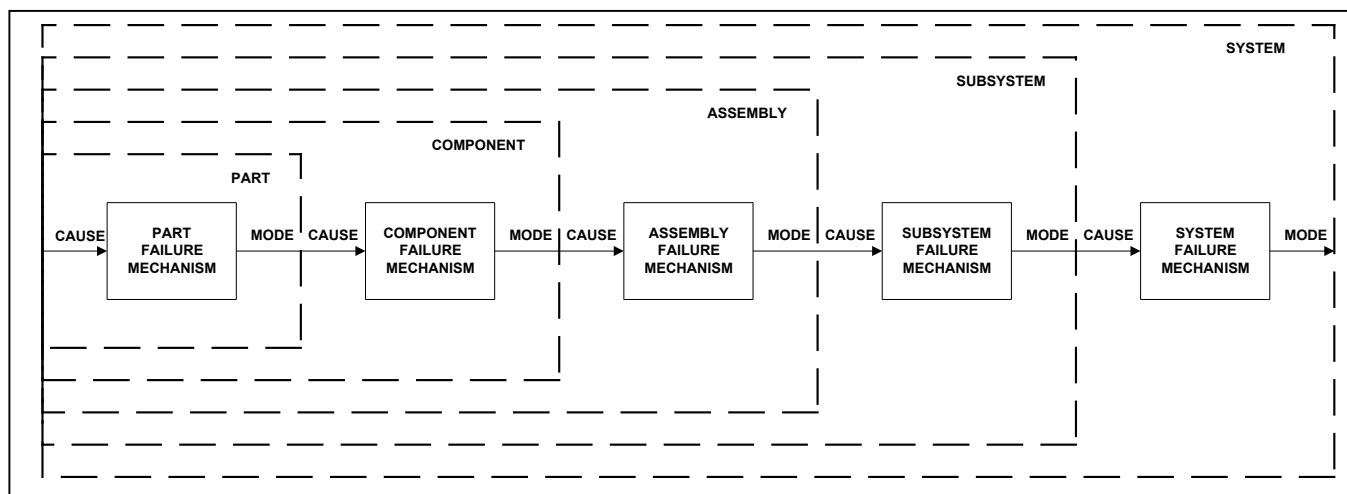


Figure 1 — Failure Process Flow

The contractor shall collect sufficient system design information to identify and evaluate inherent, functional and physical, singular failure modes. If a Capability Level 2 or higher product FMECA is required, the contractor shall collect sufficient system design information to identify and evaluate inherent and induced, functional and physical, singular, sequential, and combinational failure modes. The types of induced modes to be identified and evaluated shall include over-stressed items (see Reference 10), out-of-tolerance functions (see Reference 11), latent design concerns and drawing errors (see Reference 9), sneak circuits (see Reference 8), the end-effects of manufacturing, operator, and maintenance errors (see Reference 18), extreme environment electronics effects (see Reference 21), human errors (see Reference 12), and software faults (see Reference 17). Reference 23 provides examples of inherent and induced failure causes; Reference 24 provides a list of sneak circuit clues; and Reference 25 provides guidelines for performing design concern and drawing error analysis.

The severity classification for each postulated failure mode shall be defined and qualified or rated based on the specific effect(s) of the failure mode on the system/mission. Table 1 provides the baseline failure mode severity classification criteria. All item failure mode and system level failure effect information shall be included in the product FMECA evaluation. Where a single item failure mode is found to be non-critical or non-detectable, the analysis shall be extended to identify possible catastrophic or critical effects due to a non-critical or non-detectable sequential⁶ or concurrent⁷ failure mode contributed by other failed items. When safety, redundant, or back-up items exist, the analysis shall be broadened to include the failure conditions that resulted in the need for the safety, redundant, or back-up item. If a Capability Level 3 or higher product FMECA is required, the FMEA shall be documented in the product FMECA/hazards analysis database. All failure modes that pose an unacceptable risk shall be uniquely identified in the product FMECA to maintain their visibility for proper handling.

Table 1 — AIAA S-102 Failure Severity Classification

Failure Severity Classification	Failure Effect Description
CATASTROPHIC	1) failure would cause loss of life or total disability to personnel, or 2) failure would cause identifiably catastrophic damage to system and repairs that are beyond the capability of the user or contractor to resolve the effects.
CRITICAL	1) failure would cause severe disabling injury or severe occupational illness to personnel, or 2) failure would cause identifiably critical damage to the system and extensive repairs to resolve the effects.
MARGINAL	1) failure would cause minor injury or minor occupational illness to personnel that may require hospitalization, but failure is not disabling, or 2) failure would cause identifiably marginal damage to the system and acceptable level of repairs and downtime to resolve effects.
MINOR	1) failure would cause minor injury to personnel but those injuries would not require hospitalization, or failure would cause minor occupational illness, or 2) failure would cause identifiably minor damage to the system and minor repairs and short downtime to resolve effects.
NEGLIGIBLE	1) failure would cause less than minor injury and no occupational illness, or 2) failure would cause negligible damage to the system and insignificant or no downtime to resolve effects, or 3) failure is not credible.

⁶ For this standard, a sequential failure mode means the same as a cascading failure mode.

⁷ For this standard, a concurrent failure mode means the same as multiple failure modes.

5.4 Criticality Analysis

The contractor shall perform a criticality analysis (CA). For this standard, CA is defined as the grouping of all failure modes according to severity classification and the estimating of the probability of occurrence for each failure mode. Critical failure modes that are credible shall be evaluated further for the appropriate failure compensation methods. Critical failure modes that are non-credible shall be identified. If a Capability Level 3 or higher product FMECA is required, the CA shall be documented in the product FMECA/hazards analysis database. Guidance for performing a CA is provided in Reference 23.

5.5 Failure Detection Analysis

The contractor shall identify the system features that are available to detect each failure mode (see Reference 13) and probability of success for each detection action. When sufficient design information is available, the signal failure signatures at the appropriate failure detection test points shall be identified for each failure mode. All mission critical failure modes that have a 0% to less than 99% chance of being successfully detected within the time required shall be reported to project management and have their sources placed under critical item control. If a Capability Level 3 or higher product FMECA is required, the failure detection analysis shall be documented in the product FMECA/hazards analysis database to the extent of allowing independent verification of the system's failure detection capability.

5.6 Failure Isolation Analysis

If a Capability Level 4 or higher product FMECA is required, the contractor shall identify the system features that are available to isolate each failure mode (see Reference 13), and the ambiguity level for each isolation action. When sufficient design information is available, the signal failure signatures at the appropriate failure isolation test points shall be evaluated for each failure mode. If a Capability Level 3 or higher product FMECA is required, the failure isolation analysis shall be documented in the product FMECA/hazards analysis database to the extent of allowing independent verification of the system's failure isolation capability.

5.7 Detection Risk Priority Analyses

The contractor shall perform a detection risk priority analysis (RPA). If a Capability Level 3 or higher product FMECA is required, the RPA shall be documented in the product FMECA/hazards analysis database.

5.8 Reliability, System Safety, and Maintainability Critical Item Analyses

The contractor shall establish processes for identifying, analyzing, and controlling R&M critical items in a timely manner (see Reference 3). If a Capability Level 2 or higher product FMECA is required, the contractor shall establish processes for identifying, analyzing, and controlling system safety and maintainability critical items in a timely manner (see Reference 3). The contractor shall describe the approach for identifying, analyzing, and controlling critical items in the product FMECA plan.

5.9 Failure Compensation Method Analysis

The contractor shall identify the methods considered for compensating the effects of each unacceptable credible failure mode, or provide rationale for retaining an unacceptable credible failure mode in the design. The failure compensation methods considered shall be described in the product FMECA report along with their respective advantages and disadvantages. At least one failure compensation method (e.g., design change, manufacturing process change, special quality control measure, special test, or operational corrective action) shall be considered for each catastrophic (Severity Classification 5) and critical (Severity Classification 4) failure mode that is judged to have a credible likelihood of occurrence during the prescribed mission. Each decision to select a particular failure compensation method shall be documented in the product FMECA report. The contractor shall implement and verify the selected failure compensation methods using processes established for critical item control.

5.10 Integrated Product FMECA/Hazards Analysis Database

If a Capability Level 3 product FMECA is required, the contractor shall establish an integrated product FMECA/hazards analysis database that contains the FMECA data products that are identified in the established systems engineering data flow schemas for all applicable product development phases and has data change control and tracking procedures⁸. The database shall document the results of processes used to identify, analyze, track, and control failure modes and system safety hazards that pose unacceptable risk. The combined FMECA/hazard analysis database shall include a description of the failure mode or safety hazard, identify the site and causal events or factors, and assign a criticality number or hazard risk number to the identified item, along with corrective action status and required level of authority to accept the failure mode or safety hazard. All data that are entered in or extracted from the product FMECA/hazards analysis database shall be prefaced with one or more keyword data element descriptions (DED) listed in Annex C. Each keyword DED belongs to one of the following data types:

- Physical or Functional Characteristic
- Physical or Functional Dependency
- Application
- Failure Mode and Effects Analysis
- Criticality Analysis
- Maintainability Analysis
- Anomaly Detection, Isolation, and Response
- Reliability, System Safety, and Maintainability Critical Item
- Hazards Analysis
- Failure/Hazard Compensation
- Identification
- Unit
- Reference
- Event
- Diagnostic
- Value
- Comment
- Attachment
- Database Administration

⁸ The objective here is to ensure that all failure modes, safety hazards, and critical items are documented; the history of designed-in reliability and safety improvements is maintained; and current data is distinguishable from out-of-date data.

The product FMECA/hazards analysis database shall be structured to allow:

- a) independent verification of the system level effects for all postulated failure modes that are Severity Classification 3, 4, or 5, and
- b) online review of the most current and all prior product FMECA reports.

If a Capability Level 4 product FMECA is required, the contractor shall establish and maintain a seamless interface between the product FMECA/hazards analysis database and the project R&M database system (see Reference 4). For this latter case, all product FMECA datasets and reports shall be retrievable from the project R&M database system.

5.11 Data Exchange Between Product FMECA Process And Other Project Functions

If a Capability Level 3 or higher product FMECA process is required, the contractor shall assure timely utilization of the product FMECA database by project functions, such as, system safety, test, quality assurance, and risk management, and R&M program tasks, such as, system reliability modeling (see Reference 6), maintainability predictions (see Reference 14), and reliability development growth testing (see Reference 22), to aid in qualifying the severity of broad categories of system deficiencies as part of the overall mission assurance effort. The contractor shall establish the product FMECA process as an integral part of the product life cycle systems engineering process, from the early design phase of system definition through fabrication, assembly, integration, and test (FAIT). The exchange of product FMECA data products shall be based on the established systems engineering data flow schemas for all applicable product development phases. If a Capability Level 4 or higher product FMECA process is required, all FMECA data that are exchanged in the project shall be prefaced with one or more keyword data element descriptions (DED) listed in Annex C. The contractor shall make every effort to utilize product FMECA data to avoid duplication of effort whenever possible, and continuously update the product FMECA as design changes occur and “real world” engineering information⁹ becomes available.

5.12 Product FMECA Process Performance Evaluation

If a Capability Level 4 or higher product FMECA process is required, the contractor shall take measures to continuously evaluate and improve the product FMECA process. This evaluation shall include the following tasks:

- a) Evaluate and document the value of the completed product FMECA to internal organizations or projects by surveying recipients of the data products, such as, design, test, and operations. Use the results of the survey questionnaire to identify which recipients of the data products require aid to realize its full value for specific uses;
- b) Quantify or rate the consequences of each system-level functional failure based on the specific effect(s) of loss of the function on the mission;
- c) Quantify or rate the likelihood that accurate input data are collected and applied, given a specific data risk classification, as defined by the set of input data maturity rating criteria in Table 2. This activity provides process validation in so far as it estimates the “goodness” of the input data;
- d) Quantify or rate the likelihood that the appropriate failure compensation methods are identified and applied, given a specific decision risk classification, as defined by the set of failure compensation

⁹ FRACAS is a major source of “real-world” engineering information. “Real-world” engineering information might uncover an overlooked SPFM item or indicate that a non-credible SPFM item should be reclassified as an unacceptable failure mode (i.e., likelihood or uncertainty of failure is greater than 1.0E-6).

method rating criteria in Table 3. This activity provides process validation in so far as it estimates the “goodness” of the failure compensation methods; and

- e) Summarize the product FMECA process risk in an appropriate format, such as, the FMECA dataset maturity versus failure mode severity matrix provided as Figure 2. Note that the Y-axis of the matrix rates the severity of the failure’s impact, and the X-axis of the matrix rates the maturity of the selected FMECA dataset.

The R&M program shall set annual improvement goals for the product FMECA process, in consultation with the project and institutional management. The improvement goals shall be directed at achieving a level of product FMECA process maturity that is consistent with the level of residual risk acceptable to the project. This residual risk results from the potential of not achieving specific mission objectives if the product FMECA were to be performed inappropriately.

5.12.1 Product FMECA Dataset Appropriateness Rating

If a Capability Level 4 or higher product FMECA is required, the contractor shall rate the maturity of each FMECA dataset in respect to appropriateness according to the criteria in Table 2.

Table 2 — Product FMECA Dataset Maturity Rating Criteria

Maturity Level 5	Maturity Level 4	Maturity Level 3	Maturity Level 2	Maturity Level 1
Identification of detailed failure modes is based on detailed simulation modeling or diagnostics	Identification of detailed failure modes is based on detailed static modeling	Identification of generalized failure modes is based on approximate simulation modeling	Identification of generalized failure modes is based on approximate static modeling	Identification of generalized failure modes is based on experience instead of modeling

5.12.2 Selected Failure Compensation Method Appropriateness Rating

If a Capability Level 4 or higher product FMECA is required, the contractor shall rate the maturity of each selected failure compensation method in respect to the appropriateness of the selected failure compensation methods according to the criteria in Table 3.

Table 3 — Failure Compensation Method Maturity Rating Criteria

Maturity Level 5	Maturity Level 4	Maturity Level 3	Maturity Level 2	Maturity Level 1
1) Information believed necessary for investigating “root cause” is identified and documented in FMECA 2) The “definite” root cause(s) is identified 3) Contractor approves a comprehensive failure compensation feature that is based on appropriate technical or statistical analyses that fully address the “definite” root cause(s) mitigation	1) Information believed necessary for investigating “root cause” is identified but only some of that information is documented in FMECA 2) As a result, the “candidate” root cause(s) is identified instead of the “definite” root cause 3) Contractor approves a comprehensive failure compensation feature that is based on appropriate technical or statistical analyses that fully address the “candidate” root cause(s) mitigation	1) Information believed necessary for investigating “root cause” is identified but only some of that information is documented in FMECA 2) As a result, the “candidate” root cause(s) is identified instead of the “definite” root cause 3) Contractor approves a limited failure compensation feature that is based on appropriate technical or statistical analyses, but the analyses do not fully address the “candidate” root cause(s) mitigation	1) Information believed necessary for investigating “root cause” is identified but only some of that information is documented in FMECA 2) As a result, the “candidate” root cause(s) is identified instead of the “definite” root cause 3) Contractor approves “use as is” disposition of unacceptable failure mode based on inappropriate technical or statistical analysis	1) Contractor ignores the “definite” root cause even though identification approach is based on appropriate technical or statistical analysis method 2) Instead, contractor approves “use as is” disposition of unacceptable failure mode based on inappropriate technical or statistical analysis method

5.12.3 Product FMECA Process Performance Evaluation

If a Capability Level 4 or higher product FMECA is required, the contractor shall group the FMECA datasets according to their maturity rating and severity classification¹⁰, and group the selected failure compensation methods according to their maturity rating and severity classification.

The minimally acceptable thresholds for the data maturity ratings shall comply with Figure 2 and Figure 3. The contractor shall use the results of the product FMECA performance evaluation to aid in predicting anomalies during system test and operation (STO).

5.13 Lessons Learned

If a Capability Level 3 product FMECA process is required, the contractor shall review existing FMECA/Hazards data/reports for applicable product-based¹¹ and process-based¹² lessons learned. The candidate lessons learned shall be evaluated for quality, prioritized, and forwarded to the Lessons Learned Approval Authority for appropriate action. The contractor shall take steps to ensure that candidate lessons learned are documented and reviewed in a timely manner, and the related recommendations infused throughout the project, the stakeholder organizations, and as necessary, enterprise-wide using the appropriate system. The product FMECA/hazards analysis database shall

¹⁰ The basic premise here is that the selected failure compensation method can be only as good as its associated FMECA dataset and selection rationale.

¹¹ For this standard, a product-based lesson learned is important information created, documented, and retrieved according to a system or device life cycle specific functional or physical descriptor.

¹² For this standard, process-based lesson learned is important information created, documented, and retrieved according to a process or procedure descriptor.

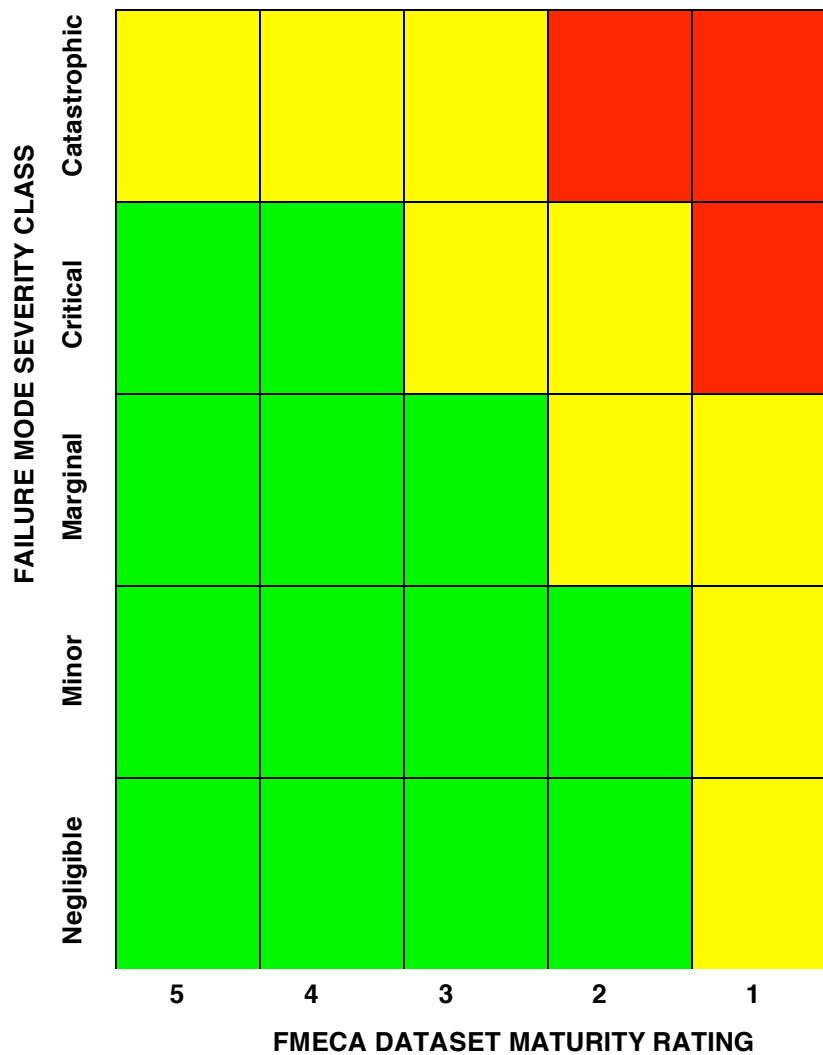
include a field that permits an authorized person to indicate that particular data is a lessons learned candidate. A positive indication in the product FMECA/hazards analysis database lessons learned field shall generate a notification to a Lessons Learned Review Committee or similar approval authority regarding the data's candidacy.

If a Capability Level 4 product FMECA process is required, the contractor shall exchange product FMECA lessons learned with other projects throughout the enterprise, and shall review the FMECA lessons learned that are received from other projects to identify needed process improvements, such as, improved procedures or training materials.

If a Capability Level 5 product FMECA process is required, the contractor shall exchange non-proprietary lessons learned data with other organizations, entering into agreements and employing safeguards to protect security-classified, International Traffic in Arms Regulations (ITAR)-restricted, proprietary, or other sensitive data. The received data shall be reviewed by an enterprise-level Lessons Learned Board to identify significant recommendations that should be implemented on the project or enterprise-wide.

5.14 Structured Review

If a Capability Level 5 or higher FMECA is required, the contractor shall develop and apply a structured review process to aid participation in and evaluation of the product FMECA process at all life cycle development phases. The review process shall include personnel who are cognizant of events that led to design errors, design weaknesses, and test failures in systems similar to the one being developed. Product-based and process-based lessons learned that are relevant to the system being developed shall be gathered across the enterprise and used to develop review checklists that support timely implementation of the structured review process and updating of the FMECA. The review checklists shall reflect the technical knowledge, insights, design rules, application data, and other clues that helped uncover design errors and design weaknesses in systems similar to the one being developed.

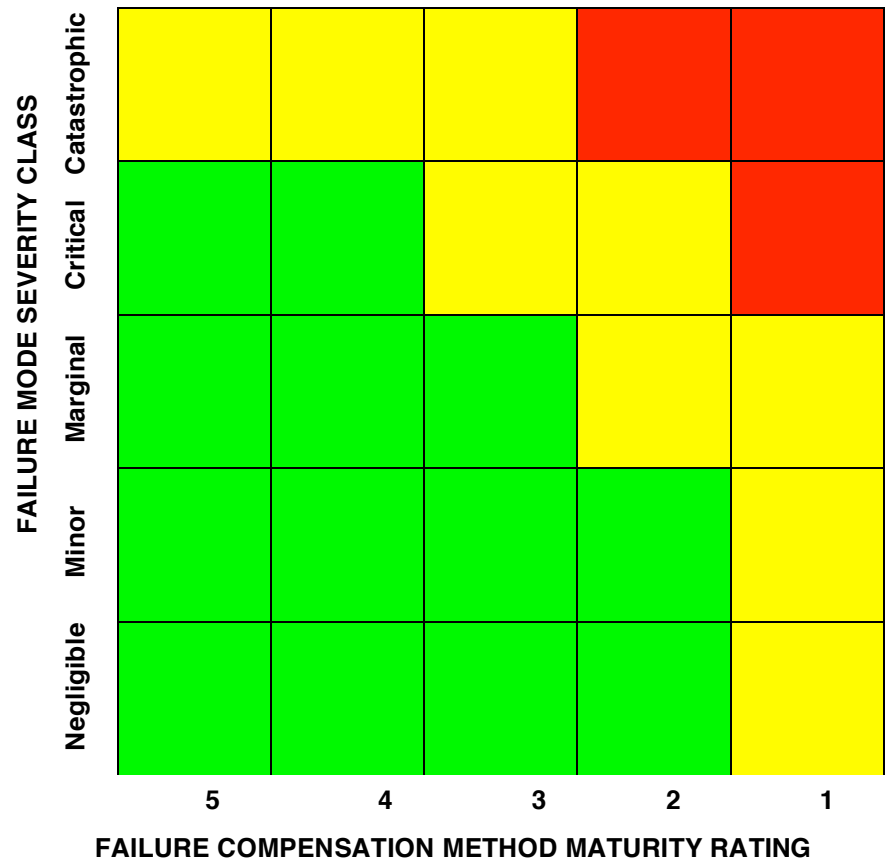


Legend

- FMECA dataset maturity is acceptable.
- FMECA dataset maturity may be marginally acceptable (may warrant watch item status).
- FMECA dataset maturity is not acceptable.

NOTE The minimum acceptable thresholds are shown.

Figure 2 — Product FMECA Dataset Evaluation



Legend



Failure compensation method maturity is acceptable.



Failure compensation method maturity is marginally acceptable (may warrant watch item status).



Failure compensation method maturity is not acceptable.

NOTE The minimum acceptable thresholds are shown.

Figure 3 — Failure Compensation Method Evaluation

Annex A AIAA S-102 Document Tree (normative)**Performance-Based Reliability & Maintainability Program Requirements****S-102.0 Performance-Based Reliability & Maintainability Program General Requirements**

S-102.1 Performance-Based R&M Management Requirements		S-102.2 Performance-Based R&M Engineering & Analysis Requirements		S-102.3 Performance-Based R&M Testing Requirements	
S-102.1.1	R&M Program Planning	S-102.2.1	Functional Dependency & Architecture Modeling	S-102.3.1	Environmental Stress Screening
S-102.1.2	Subcontractor and Supplier Monitoring & Control	S-102.2.2	System Reliability Modeling	S-102.3.2	Reliability Development Growth Testing
S-102.1.3	R&M Program Working Groups & Reviews	S-102.2.3	Component Reliability Predictions	S-102.3.3	Reliability, Maintainability, and Availability Demonstration Testing
S-102.1.4	Failure Reporting, Analysis, and Corrective Action System	S-102.2.4	Product Failure Mode, Effects, and Criticality Analysis	S-102.3.4	Reliability Life Testing
S-102.1.5	Failure Review Board	S-102.2.5	Sneak Circuit Analysis	S-102.3.5	Design of Tests
S-102.1.6	R&M Critical Item Risk Management	S-102.2.6	Design Concern & Drawing Error Analysis		
S-102.1.7	Project R&M Database System	S-102.2.7	Parts Stress Derating Analysis		
		S-102.2.8	Worst Case Analysis		
		S-102.2.9	Human Error Predictions		
		S-102.2.10	Environmental Event Survivability Prediction		
		S-102.2.11	Anomaly, Detection, and Response Analysis		
		S-102.2.12	Maintainability Predictions		
		S-102.2.13	Operational Dependability and Operational Availability Modeling		
		S-102.2.14	Hazard Analysis		
		S-102.2.15	Software Reliability Predictions		
		S-102.2.16	Process Failure Mode, Effects, and Criticality Analysis		
		S-102.2.17	Event Tree Analysis		
		S-102.2.18	Fault Tree Analysis		
		S-102.2.19	Fishbone Analysis		
		S-102.2.20	R&M Similarity and Allocations Analysis		
		S-102.2.21	Parts Engineering and Analysis		
		S-102.2.22	Stress and Damage Simulation Analysis		

Annex B AIAA S-102 Product FMECA Capability Level Requirements (normative)

B.1 The Capability Level 1 Product FMECA shall include the following tasks:

B.1.1 Timely establishment of the requirements and analytical ground rules for the product FMECA (this is a process validation activity when it includes evaluation of the appropriateness of the FMECA process prior to its use);

B.1.2 Timely establishment of product FMECA technical performance metrics (TPM);

B.1.3 Timely collection and evaluation of the necessary system design information to identify and eliminate or control all inherent, singular H/W failure modes that pose unacceptable risk across the system life cycle. The scope of the system design information that is collected and evaluated shall include the following:

- All inherent¹³, functional and physical, singular H/W failure modes (within analytical ground rules to be specified by the contractor);
- The effects of each inherent, functional or physical, singular H/W failure mode on every system level;
- The detection methods and associated detection success rates for all inherent, functional and physical, singular H/W failure modes;

B.1.4 Timely creation of a product FMECA dataset for each postulated system failure mode;

B.1.5 Timely application of reliability critical Item (RCI) selection criteria and the evaluation of RCI failure modes;

B.1.6 Timely calculation of the criticality number (CN) and detection risk priority number (RPN), which is the product of CN and the detection probability of failure, for each inherent, functional or physical, singular H/W failure mode (within analytical ground rules to be specified by the contractor). The CN and RPN analytical ground rules shall require numerical estimates for all credible Severity Classification 3 or higher failure modes;

B.1.7 Timely identification of scheduled project activities that are intended to help reduce the likelihood, or manage the consequences of the following:

- Each credible RCI;
- Each credible, Severity Classification 3 or higher, inherent, functional or physical, singular H/W failure mode;

B.1.8 Timely identification and documentation of all non-credible Severity Classification 3 or higher inherent, functional and physical, singular H/W failure modes;

B.1.9 Timely validation of each product FMECA dataset;

B.1.10 Timely development and documentation of a product FMECA report that describes the system's intended uses, functional dependencies, operating modes, and the FMECA data products.

¹³ For this standard, inherent failure modes are the result of characteristic weaknesses related to an item's specified design and materials.

B.2 The Capability Level 2 Product FMECA shall include all the tasks in the Capability Level 1 Product FMECA plus the following:

B.2.1 Timely development, documentation, and flow down, as appropriate, of a product FMECA process plan that is based on industry-accepted concepts for performance-based practices and is an integral part of the R&M program plan. The product FMECA process plan shall describe the objectives, analytical ground rules or scope, assumptions, activities or approach, performance metrics, data products, and the organizational elements responsible for generating and processing the product FMECA data products;

B.2.2 Timely collection and evaluation of the necessary system design and processing information to identify and eliminate or control all induced¹⁴, functional and physical, singular H/W failure modes that pose unacceptable risk across the system life cycle. The scope of the system design information that is collected and evaluated shall include the following:

- All induced, functional and physical, singular H/W failure modes (within analytical ground rules to be specified by the contractor);
- The effects of each induced, functional or physical, singular H/W failure mode on every system level;
- The detection methods and associated detection success rate for all induced, functional and physical, singular H/W failure modes;

B.2.3 Timely application of system safety critical item (SSCI) and maintenance critical item (MCI) selection criteria, and evaluation of SSCI and MCI failure modes;

B.2.4 Timely calculation of the criticality number (CN) and detection risk priority number (RPN), which is the product of CN and the detection probability of failure, for each induced, functional or physical, singular H/W failure mode (within analytical ground rules to be specified by the contractor). The CN and detection RPN analytical ground rules shall require numerical estimates for all credible Severity Classification 3 or higher failure modes;

B.2.5 Timely identification of failure mechanisms and root causes for each Severity Classification 3 or higher inherent or induced, functional or physical, singular H/W failure mode;

B.2.6 Timely identification of special design features or failure mitigation actions that are intended to eliminate, reduce the likelihood, or manage the consequences of the following:

- Each credible SSCI and MCI;
- Each credible, Severity Classification 3 or higher, inherent or induced, functional or physical, singular H/W failure mode;

B.2.7 Timely identification and documentation of all non-credible Severity Classification 3 or higher induced, functional and physical, singular H/W failure modes.

B.3 The Capability Level 3 Product FMECA shall include all the tasks in the Capability Level 2 Product FMECA plus the following:

B.3.1 Timely collection and evaluation of the necessary system design and processing information to identify and eliminate or control all inherent and induced, functional and physical, 2nd-order sequential and combinational, H/W and S/W failure mode sets that pose unacceptable risk across the system life

¹⁴ Induced failure modes include operator or maintenance procedure errors, malicious actions, and design oversights that result in electrical or thermal overstress, out-of-tolerance conditions, latent design concerns, drawing errors, and sneak circuits.

cycle. The scope of the system design information that is collected and evaluated shall include the following:

- All inherent and induced, functional and physical, 2nd-order sequential and combinational, H/W and S/W failure mode sets (within analytical ground rules to be specified by the contractor);
- The effects of each inherent or induced, functional or physical, 2nd-order sequential or combinational, H/W and S/W failure mode set on every system level;
- The detection methods and associated detection success rates for each inherent or induced, functional or physical, 2nd-order sequential or combinational, H/W and S/W failure mode set;

B.3.2 Timely calculation of the criticality number (CN) and detection risk priority number (RPN), which is the product of CN and the detection probability of failure, for each inherent or induced, functional or physical, 2nd-order sequential or combinational, H/W and S/W failure mode set (within analytical ground rules to be specified by the contractor). The CN and detection RPN analytical ground rules shall require numerical estimates for all credible Severity Classification 3 or higher failure modes;

B.3.3 Timely identification of failure mechanisms and root causes for each Severity Classification 3 or higher inherent or induced, functional or physical, 2nd-order sequential or combinational, H/W and S/W failure mode set;

B.3.4 Timely identification of special design features or failure mitigation actions that are intended to eliminate, reduce the likelihood, or manage the consequences of each credible, Severity Classification 3 or higher, inherent or induced, functional or physical, 2nd-order sequential or combinational, H/W and S/W failure mode set;

B.3.5 Timely identification and documentation of all non-credible Severity Classification 3 or higher inherent and induced, functional and physical, 2nd-order sequential and combinational, H/W and S/W failure mode sets;

B.3.6 Timely development and maintenance of an integrated product FMECA/hazards analysis database that (1) allows checking inherent and induced, functional and physical, singular H/W failure modes, and 2nd-order sequential and combinational, H/W and S/W failure mode sets against verified design drawings, (2) automatically generates the product FMECA report, and (3) permits an authorized person to indicate particular data is a lessons learned candidate;

B.3.7 Timely utilization of the product FMECA/hazards analysis database to the greatest extent practical by project activities, such as, system safety and logistics, and R&M program tasks, such as, FRACAS, system reliability modeling, and maintainability predictions, to aid in qualifying the severity of broad categories of system deficiencies as part of the overall mission assurance effort;

B.3.8 Timely collection and review of existing product FMECA lessons learned that are: (1) derived from sources internal to the enterprise, and (2) relevant to the system being developed. The objective of this activity is to identify needed FMECA process improvements;

B.3.9 Timely evaluation of all aspects of the product FMECA process, including its implementation and data products, to identify candidate product-based and process-based candidate lessons learned. Evaluate for quality and prioritize these lessons learned, and forward them to the Lessons Learned Approval Authority (LLAA) for appropriate action;

B.3.10 Timely development and documentation of sufficient detailed information in the product FMECA report to allow independent evaluation of the comprehensiveness of the analysis.

B.4 The Capability Level 4 Product FMECA shall include all the tasks in the Capability Level 3 Product FMECA plus the following:

B.4.1 Timely collection and evaluation of the necessary system design and processing information to identify the isolation methods for the failure sites of all inherent and induced, functional and physical, singular H/W failure modes, and 2nd-order sequential and combinational, H/W and S/W failure mode sets;

B.4.2 Timely development and maintenance of a product FMECA/hazards analysis database that is a part of, is integrated with, or interfaces with a project R&M database that complies with S-102 R&M keyword data element description (DED) requirements;

B.4.3 Timely evaluation of the product FMECA dataset in accordance with the project risk management plan and the dataset maturity criteria defined in this standard (this is a process verification activity in so far as it verifies that the appropriate methods were used to create the FMECA dataset);

B.4.4 Timely evaluation and documentation of the value of the completed product FMECA to internal organizations or projects by surveying recipients of the data products, such as, design, test, and operations. Use the results of the survey questionnaire to identify which recipients of the data products require aid to realize its full value for specific uses;

B.4.5 Evaluate aspects of the product FMECA process, and its implementation, for valid lessons learned candidates. Submit the lesson candidates to the Lessons Learned Approval Authority (LLAA) for review. Exchange approved lessons with other projects throughout the enterprise. Review lessons received from other projects to identify product FMECA process improvements that should be implemented, such as, improvements in automated tools or training materials.

B.5 The Capability Level 5 Product FMECA shall include all the tasks in the Capability Level 4 Product FMECA plus the following:

B.5.1 Timely development and implementation of a structured review process¹⁵ (independent review team/panel, independent technical assessment, etc.) that draws heavily on the product FMECA experiences of other projects to aid the implementation or evaluation of the product FMECA process of concern, as appropriate, in all product life cycle development phases;

B.5.2 Timely establishment of a process for the continuous improvement of the enterprise-approved documented product FMECA practices and training materials. This activity includes annual goal setting and periodic independent evaluations of the organization's progress toward those goals;

B.5.3 Timely collection and review of product FMECA lessons learned that are documented by outside organizations to identify significant recommendations that should be implemented by the project. Share FMECA lessons learned that are not subject to proprietary or legal constraints with external organizations through established channels, such as, the Government-Industry Data Exchange Program (GIDEP) or a non-profit research and development (R&D) consortium. (This activity requires establishing appropriate safeguards for security-classified, ITAR-restricted, and proprietary data.)

¹⁵ The development and implementation of a comprehensive review checklist facilitates a structured review process.

Annex C AIAA S-102 Product FMECA Keyword Data Element Descriptions (normative)

XML Data Exchange Keywords (case sensitive)	FMECA Report Keywords (not case sensitive)	Data Type
FIRST_ORDER_FM	1st order failure mode	FMEA
SECOND_ORDER_FM	2nd order failure mode	FMEA
THIRD_ORDER_FM	3rd order failure mode	FMEA
FOURTH_ORDER_FM	4th order failure mode	FMEA
ACCESSIBILITY	Accessibility	Maintainability analysis
ACCIDENT	Accident	FMEA
ACTION_STATUS	Action status	Database administration
ALDT	Administrative and logistics delay time	FMEA
ALIGN	Align	Event
ALIGNMENT	Alignment	Characteristic
ALLOCATED	Allocated	Reference
ALLOCATION	Allocation	Reference
AMBIGUITY	Ambiguity	Diagnostic
AMBIGUITY_GROUP	Ambiguity group	Diagnostic
AMPLITUDE	Amplitude	Characteristic
AMP	Amp	Unit
ANOMALY	Anomaly	Characteristic
AREA	Area	Characteristic
ASSEMBLY	Assembly	Identification, Event
ASSUMPTION	Assumption	FMEA
ATTACHMENT	Attachment	Attachment
ATTENUATION	Attenuation	FMEA
ATTRIBUTE	Attribute	Characteristic
AVAILABILITY	Availability	Characteristic
AVERAGE	Average	Value
BLOCK	Block	Dependency
BLOCK_DIAGRAM	Block diagram	Dependency
BROKE	Broke	FMEA
BROKEN	Broken	FMEA
BURN	Burn	FMEA
BURNED	Burned	FMEA
BURST	Burst	FMEA
CABLE	Cable	Characteristic
CAPACITANCE	Capacitance	Characteristic
CASCADING_FAILURE_MODE	Cascading failure mode	FMEA
CAUSE	Cause	FMEA
CELSIUS	Celsius	Unit
CHARGE	Charge	Characteristic
CHARGED_PARTICLE	Charged particle	Characteristic
CHARACTERISTIC	Characteristic	Characteristic

XML Data Exchange Keywords (case sensitive)	FMECA Report Keywords (not case sensitive)	Data Type
CHEMICAL	Chemical	Characteristic
CHILD	Child	Dependency
CIRCUIT	Circuit	FMEA
CLEAN	Clean	Characteristic
CLEANED	Cleaned	Event
CLOGGED	Clogged	FMEA
COLLIDE	Collide	FMEA
COLLISION	Collision	FMEA
COLOR	Color	Characteristic
COMMAND	Command	Event
COMMANDED	Commanded	Event
COMPENSATING_PROVISION	Compensating provision	Failure compensation
COMPONENT	Component	Identification
CONDITIONAL_PROBABILITY	Conditional probability	FMEA
CONDUCTED	Conducted	Event
CONDUCTED_INTERFERENCE	Conducted interference	FMEA
CONFIGURATION	Configuration	Characteristic
CONFIGURATION_ITEM	Configuration item	Characteristic
CONNECTOR	Connector	Characteristic
CONNECTOR_PIN	Connector pin	Characteristic
CONTAMINATED	Contaminated	FMEA
CONTAMINATION	Contamination	FMEA
CONTROL	Control	Characteristic
CONTROL_LEADS	Control leads	Characteristic
COORDINATES	Coordinates	Characteristic
CORRECTIVE_ACTION	Corrective action	Failure compensation
CORROSION	Corrosion	FMEA
COSMIC_RAYS	Cosmic rays	FMEA
CRACK	Crack	FMEA
CRITICAL_ITEM	Critical item	Critical item
CRITICALITY	Criticality	Criticality analysis
CURRENT	Current	Characteristic
DAMAGE	Damage	FMEA
DAMAGED	Damaged	FMEA
DATA_MATURITY	Data maturity	Reference
DATA_SOURCE	Data source	Reference
DAY	Day	Unit
DELAYED	Delayed	FMEA
DEPENDABILITY	Dependability	Characteristic
DESIGN	Design	Event, Identification
DESIGN_CONCERN	Design concern	FMEA
DESIGN_CONCERN_CLUE	Design concern clue	Reference

XML Data Exchange Keywords (case sensitive)	FMECA Report Keywords (not case sensitive)	Data Type
DESIGN_ERROR	Design error	FMEA
DESIGN_PROVISION	Design provision	Failure compensation
DETECTABILITY	Detectability	Diagnostic
DETECTION	Detection	Diagnostic
DIMENSIONS	Dimensions	Characteristic
DIRTY	Dirty	Characteristic
DISCHARGE	Discharge	Event, FMEA
DISCHARGED	Discharged	Event, FMEA
DISTRIBUTED	Distributed	Characteristic
DISTRIBUTION	Distribution	Characteristic
DOCUMENT	Document	Reference
DOCUMENTED	Documented	Characteristic
DORMANT	Dormant	Characteristic
DORMANT_TIME	Dormant time	Characteristic
DRAWING_ERROR	Drawing error	FMEA
DURABILITY	Durability	Characteristic
EARLY	Early	FMEA
EFFECTS	Effects	FMEA
ELECTRICAL	Electrical	Characteristic
ELECTRICAL_INTERFERENCE	Electrical interference	FMEA
ELECTRONIC	Electronic	Characteristic
END	End	Database management
ELECTROMAGNETIC COMPATIBILITY	Electromagnetic compatibility	FMEA
ELECTROMAGNETIC INTERFERENCE	Electromagnetic interference	FMEA
ELECTROMECHANICAL	Electromechanical	Characteristic
END_EFFECTS	End effects	FMEA
ERROR	Error	FMEA
EMLC	Errors per million lines of code	Unit
ENVIRONMENT	Environment	Application
EVENT	Event	Reference
FAILURE	Failure	FMEA
FAILURE_AMBIGUITY	Failure ambiguity	Diagnostic
FAILURE_CAUSE	Failure cause	FMEA
FAILURE_DATA	Failure data	Criticality analysis
FAILURE_DETECTION	Failure detection	Diagnostic
FAILURE_DETECTION_METHOD	Diagnostics method	Diagnostic
FAILURE_DETECTION_PROBABILITY	Diagnostics probability	Diagnostic
FAILURE_EFFECT	Failure effect	FMEA
FAILURE_ISOLATION	Failure isolation	Diagnostic
FAULT_DETECTION	Fault detection	Diagnostic
FAULT_ISOLATION	Fault isolation	Diagnostic
FAILURE_ISOLATION_METHOD	Failure isolation method	Diagnostic

XML Data Exchange Keywords (case sensitive)	FMECA Report Keywords (not case sensitive)	Data Type
FAILURE_ISOLATION_PROBABILITY	Failure isolation probability	Diagnostic
FAILURE_MODE	Failure mode	FMEA
FAILURE_MODE_RATIO	Failure mode ratio	FMEA
BITS	Failures per million hours	Unit
FITS	Failures per billion hours	Unit
FAILURE_PREDICTABILITY	Failure predictability	FMEA
FAILURE_PROBABILITY	Failure probability	FMEA
FAILURE_RATE	Failure rate	Criticality analysis
FAILURE_SEVERITY	Failure severity	FMEA
FAHRENHEIT	Fahrenheit	Unit
FATIGUE	Fatigue	FMEA
FAULT	Fault	FMEA
FAULT_COVERAGE	Fault coverage	Characteristic
FAULT_TOLERANCE	Fault tolerance	Characteristic
FEEDBACK	Feedback	Characteristic, FMEA
FEEDTHROUGH_FILTERING	Feed-through filtering	Characteristic
FIGURE	Figure	Reference
FILTER	Filter	Characteristic
FOD	Foreign Object Damage	FMEA
FRACTURE	Fracture	FMEA
FRACTURED	Fractured	FMEA
FREQUENCY	Frequency	Characteristic
FREQUENCY_DOMAIN	Frequency domain	Characteristic
FRICTION	Friction	FMECA
FROM	From	Dependency
FUNCTION	Function	Characteristic
FUNCTIONAL_BLOCK	Functional block	Dependency
FBD	Functional block diagram	Dependency
FUNCTIONAL_DIAGRAM	Functional diagram	Dependency
FUNCTIONAL_DESCRIPTION	Functional description	Characteristic
GOAL	Goal	Reference
GRAM	Gram	Unit
HANDLING_	Handling	Event
HANDLING_ERROR	Handling error	FMEA
HARD	Hard	Characteristic
HARDENED	Hardened	Characteristic
HW	Hardware	Characteristic
HAZARD	Hazard	Hazards analysis
HAZARDOUS_ITEM	Hazardous item	Hazards analysis
HAZARD_RATE	Hazard rate	Criticality analysis
HAZARD_RISK_NUMBER	Hazard risk number	Hazards analysis
HIGH	High	Value
HIGH_PRESSURE	High pressure	FMEA

XML Data Exchange Keywords (case sensitive)	FMECA Report Keywords (not case sensitive)	Data Type
HIGH_TEMPERATURE	High temperature	FMEA
HOURL	Hour	Unit
HUMAN ERROR	Human error	FMEA
HUMAN IN THE LOOP	Human in the loop	Characteristic
HUMIDITY	Humidity	FMEA
IDENTIFICATION	Identification	Identification
INDENTURE_LEVEL	Indenture level	Dependency
INDUCED	Induced	FMEA
INDUCTANCE	Inductance	Characteristic
INHERENT	Inherent	FMEA
INPUT	Input	Dependency
INSERTION_LOSS	Insertion loss	FMEA
INSPECTED	Inspected	Event
INSPECTION	Inspection	Event
ITEM	Item	Characteristic
INTERMITTENT	Intermittent	FMEA
INITIATE	Initiate	Event
INITIATED	Initiated	Event
JOULES	Joules	Unit
KILOGRAM	Kilogram	Unit
KILOLITER	Kiloliter	Unit
KILOMETER	Kilometer	Unit
LATE	Late	FMEA
LEAD	Lead	Characteristic
LEADS	Leads	Characteristic
LEAK	Leak	FMEA
LINE	Line	Characteristic
LC	Lines of code	Unit
LOCAL_EFFECTS	Local effects	FMEA
LOCATION	Location	Characteristic
LOSE	Lose	FMEA
LOW	Low	Value
LOW_PRESSURE	Low pressure	FMEA
LOW_TEMPERATURE	Low temperature	FMEA
MAINTAINABILITY	Maintainability	Characteristic
MAINTENANCE	Maintenance	Maintainability Analysis
MAINTENANCE ERROR	Maintenance Error	Maintainability Analysis
MALICIOUS_ACT	Malicious Act	FMEA
MANUFACTURE	Manufacture	Event
MASS	Mass	Characteristic
MATURITY	Maturity	Reference
METER	Meter	Unit
MILDEW	Mildew	FMEA

XML Data Exchange Keywords (case sensitive)	FMECA Report Keywords (not case sensitive)	Data Type
MILLIGRAM	Milligram	Unit
MILLILITER	Milliliter	Unit
MILLIMETER	Millimeter	Unit
MISALIGN	Misalign	FMEA
MISTAKE	Mistake	FMEA
MISTIMED	Mistimed	FMEA
MISTIMING	Mistiming	FMEA
MISSION	Mission	Application
MOISTURE	Moisture	FMEA
MTBF	Mean time between failures	Unit
MTTF	Mean time to failure	Unit
MTTR	Mean time to repair	Unit
MECHANICAL	Mechanical	Characteristic
MONTH	Month	Unit
MULTIPLE_FAILURE_MODE	Multiple failure mode	FMEA
NAME	Name	Identification
NEXT_HIGHER_LEVEL_EFFECTS	Next higher level effects	FMEA
NEXT_HIGHER_LEVEL_FUNCTION	Next higher level function	Dependency
NEXT_HIGHER_LEVEL_ITEM	Next higher level item	Dependency
NEXT_HIGHER_LEVEL_RBD	Next higher level reliability block diagram	Criticality Analysis diagram
NOISE	Noise	FMEA
NOMINAL	Nominal	Value
NONOPERATIONAL	Non-operational	Characteristic
NONOPERATING	Non-operating	Characteristic
NONOPERATING_TIME	Non-operating time	Characteristic
OHM	Ohm	Unit
OPEN	Open	FMEA
OPENED	Opened	FMEA
OPERATIONAL	Operational	Characteristic
OPERATING	Operating	Characteristic
OPERATING_TIME	Operating time	Characteristic
OPERATIONAL	Operational	Characteristic
OPERATOR_ERROR	Operator error	FMEA
OPERATOR_PROVISIONS	Operator provisions	Failure compensation
ORPHAN	Orphan	Dependency
OUTPUT	Output	Dependency
OVER_PRESSURE	Over pressure	FMEA
OVER_SIZE	Over size	FMEA
OVERSTRESS	Overstress	FMEA
PARENT	Parent	Dependency
PART	Part	Identification
PART_NUMBER	Part number	Identification

XML Data Exchange Keywords (case sensitive)	FMECA Report Keywords (not case sensitive)	Data Type
PASSBAND_FILTER	Pass-band filter	Characteristic
PATH	Path	Characteristic
PHASE	Phase	Application
PHYSICAL	Physical	FMEA
PHYSICAL_ITEM	Physical item	FMEA
PITTED	Pitted	FMEA
PNEUMATIC	Pneumatic	Characteristic
POUND	Pound	Unit
POWER	Power	Characteristic
PROBABILITY_OF_FAILURE	Probability of failure	Criticality Analysis
PROBABILITY_OF_OCCURRENCE	Probability of occurrence	Criticality Analysis
PROBABILITY_OF_SUCCESS	Probability of success	Criticality Analysis
QUANTITY	Quantity	Characteristic
RADIO FREQUENCY	Radio Frequency	Characteristic
RFI	Radio Frequency Interference	Characteristic
REACTION	Reaction	FMEA
RECOMMENDATION	Recommendation	Failure compensation
REFERENCE	Reference	Reference
REFERENCE_DESIGNATOR	Reference designator	Identification
RELIABILITY	Reliability	Characteristic
RELIABILITY_BLOCK	Reliability block	Criticality analysis
RBD	Reliability block diagram	Criticality analysis
RELIABILITY_CONSIDERATIONS	Reliability considerations	FMEA
RELIABILITY_BLOCK	Reliability block	Criticality analysis
RELIABILITY_MODEL	Reliability model	FMEA
REMARKS	Remarks	Comment
REQUIREMENT	Requirement	Reference
RESISTANCE	Resistance	Characteristic
REWORK	Rework	Characteristic
RISK	Risk	Characteristic
RISK_PRIORITY_NUMBER	Risk priority number	Diagnostic
ROOT_CAUSE	Root cause	FMEA
SAFETY	Safety	Characteristic
SCRATCHED	Scratched	FMEA
SCOURED	Scoured	FMEA
SECOND	Second	Unit
SEVERITY	Severity	FMEA
SHAPE	Shape	Characteristic
SHARP	Sharp	FMEA
SHARP EDGE	Sharp edge	FMEA
SHOCK	Shock	FMEA
SHORT	Short	FMEA
SHORTED	Shorted	FMEA

XML Data Exchange Keywords (case sensitive)	FMECA Report Keywords (not case sensitive)	Data Type
SIGNAL	Signal	Characteristic
SOURCE	Source	Reference
SPF	Single point failure	FMEA
SPFM	Single point failure mode	FMEA
SIZE	Size	Characteristic
SLOW	Slow	FMEA
SNEAK_CIRCUIT	Sneak circuit	FMEA
SNEAK_CLUE	Sneak clue	Reference
SNEAK_CONDITION	Sneak condition	FMEA
SNEAK_INDICATOR	Sneak indicator	FMEA
SNEAK_LABEL	Sneak label	FMEA
SNEAK_PATH	Sneak path	FMEA
SNEAK_PATH_ELEMENT	Sneak path element	Dependency
SNEAK_TIMING	Sneak timing	FMEA
SW	Software	Characteristic
SWCI	Software configuration item	Characteristic
SOFTWARE_ERROR	Software error	FMEA
SW_MODULE	Software module	Characteristic
SOILED	Soiled	FMEA
SOLAR_FLARE	Solar flare	FMEA
SOURCE	Source	Reference, Dependency
SOURCE_CODE	Source code	Characteristic
SPACECRAFT_CHARGING	Spacecraft charging	FMEA
START	Start	Database management
STATIC	Static	FMEA
STATIC_CHARGE	Static charge	FMEA
STATIC_ELECTRICITY	Static electricity	FMEA
STORAGE	Storage	Characteristic
STORAGE_TIME	Storage time	Characteristic
STRAIN	Strain	FMEA
STRESS	Stress	FMEA
STRUCTURE	Structure	Characteristic
STRUCTURAL	Structural	Characteristic
STUCK_CLOSED	Stuck closed	FMEA
STUCK_OPEN	Stuck open	FMEA
SUBASSEMBLY	Subassembly	Characteristic
SUB-ITEM	Sub-item	Dependency
SUBSYSTEM	Subsystem	Identification
SUPER_BLOCK	Super block	Dependency
SUPER_ITEM	Super item	Dependency
SURFACE	Surface	Characteristic
SYSTEM	System	Identification
SYSTEM_SAFETY	System safety	Characteristic

XML Data Exchange Keywords (case sensitive)	FMECA Report Keywords (not case sensitive)	Data Type
SWITCH	Switch	Characteristic
SWITCHED	Switched	Characteristic
TABLE	Table	Reference
TEMPERATURE	Temperature	Characteristic
TERMINATE	Terminate	Event
TERMINATED	Terminated	Event
TEST	Test	Event
THERMAL_EXPANSION	Thermal expansion	Characteristic
THERMAL_RESISTANCE	Thermal resistance	Characteristic
THERMAL_RUNAWAY	Thermal runaway	FMEA
THRESHOLD	Threshold	Characteristic
THRESHOLD_EXCEEDED	Threshold exceeded	FMEA
TIGHT	Tight	FMEA
TIME	Time	Event
TIME_DOMAIN	Time domain	Characteristic
TIMING_ERROR	Timing error	FMEA
TO	To	Dependency
TRANSFER_FUNCTION	Transfer function	Characteristic
UNDER_PRESSURE	Under pressure	FMEA
UNDER_SIZE	Under size	FMEA
UNDER_STRESS	Under stress	FMEA
UNIT	Unit	Characteristic, Unit
VALUE	Value	Value
VIBRATION	Vibration	Characteristic
VOLTAGE	Voltage	Characteristic
VOLUME	Volume	Characteristic
VOLT	Volt	Unit
WEAROUT	Wearout	Characteristic
WEIGHT	Weight	Characteristic
WBS	Work breakdown structure	Dependency
WORKMANSHIP	Workmanship	Characteristic
WORN	Worn	FMEA
YEAR	Year	Unit

American Institute of Aeronautics and Astronautics

**1801 Alexander Bell Drive, Suite 500
Reston, VA 20191-4344**

www.aiaa.org

ISBN 978-1-60086-708-8

